Chapter 2. Simple Random Sampling

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1 Introduction

Definition 2.1

Simple random sampling (SRS) is the procedure in which every possible sample of size n has the same chance of being selected; in particular every population unit has the same chance of being in the sample. Without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1 - f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Motivation. In survey sampling the population is a fixed finite list and the only randomness comes from which units we select. Under simple random sampling the sample mean is unbiased, and its variance carries the finite population correction $(N - n)/N$, which vanishes when we sample everyone.

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N}(u_k - \mu)^2 \frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\begin{aligned}
\mathrm{Cov}(y_i, y_j) &= \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)}\sum_{k\neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2 \\
&= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big) \\
&= -\frac{1}{N(N-1)}\sum_k (u_k - \mu)^2 = -\frac{\sigma^2}{N-1}.
\end{aligned}$$

Variance of the mean.

$$\begin{aligned}
\mathrm{Var}(\bar{y}) &= \frac{1}{n^2}\Big[\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j}\mathrm{Cov}(y_i, y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big] \\
&= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},
\end{aligned}$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum(y_i - \bar{y})^2 = \sum(y_i - \mu)^2 - n(\bar{y} - \mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}\big[n\sigma^2 - n\mathrm{Var}(\bar{y})\big] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1}\cdot\frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\mathrm{Var}}(\bar{y})] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \mathrm{Var}(\bar{y})$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\mathrm{Cov}(y_i, y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_i y_j) = P(y_i = 1)P(y_j = 1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\mathrm{Var}(\bar{y}) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar{y} \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\mathrm{Var}}(\bar{y})} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar{y} = 16.7$, $s^2 = 81.344$.

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{s^2}{n}\cdot\frac{N-n}{N} = \frac{81.344}{10}\cdot\frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

Motivation. Often the quantity of interest is a total (total income, total harvest). Scaling the mean by N gives an unbiased estimator whose variance is N^2 times that of the mean.

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2\frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2\frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{s^2}{n} = N^2\frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788,\ 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \implies n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D+\sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

Let $y_i = 1$ if the ith unit has the characteristic and 0 otherwise. Then the population proportion is the population mean of the 0/1 values, $p = \mu$, and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - [\mathrm{Var}(\hat{p}) + p^2] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1)-(N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)}\cdot\frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6\times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400\times 0.25}{2399\times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \ \Rightarrow\ n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

What is the total cost? Pooling all 621 observations into one mean ignores the fact that the regions were sampled separately and have different population sizes. The answer is to weight each regional mean by its population size, which is the stratified estimator of Chapter 3.

8 Selected exercises

Motivation. The first item rederives the SRS variance from inclusion indicators, the device behind all unequal-probability designs; the second gives the covariance of two sample means, which Ch. 5 uses for ratio estimators; the third is a complete computation for a proportion.

Exercise 2.1 (Inclusion indicators)

Draw an SRS of size n without replacement from units $1, \dots, N$ with values $u_1, \dots, u_N$, and let $Z_k = 1$ if unit k is in the sample and $Z_k = 0$ otherwise.
(a) Show that $P(Z_k = 1) = n/N$ and, for $k \neq l$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$.
(b) Writing $\bar{y} = \frac{1}{n}\sum_{k=1}^{N} Z_k u_k$, rederive $\mathrm{E}(\bar{y}) = \mu$ and $\mathrm{Var}(\bar{y}) = \frac{S^2}{n}(1-f)$.

Solution. [Supplement]

(a) All $\binom{N}{n}$ samples are equally likely. The samples containing unit k are obtained by choosing the other $n-1$ units from the remaining $N-1$, so $P(Z_k = 1) = \binom{N-1}{n-1}/\binom{N}{n} = n/N$. Similarly the samples containing both k and l number $\binom{N-2}{n-2}$, and $\binom{N-2}{n-2}/\binom{N}{n} = \frac{n(n-1)}{N(N-1)}$.

(b)

Step 1. (Moments of the indicators.) By (a), $\mathrm{E}Z_k = f$, $\mathrm{Var}Z_k = f(1-f)$ (as $Z_k^2 = Z_k$), and for $k \neq l$

$$\mathrm{Cov}(Z_k, Z_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = \frac{n}{N}\Big(\frac{n-1}{N-1} - \frac{n}{N}\Big) = -\frac{f(1-f)}{N-1}.$$

Step 2. (Mean.) Only the Z_k are random, so $\mathrm{E}\bar{y} = \frac{1}{n}\sum_k f\,u_k = \frac{1}{N}\sum_k u_k = \mu$.

Step 3. (Variance.) Since $\sum_k Z_k = n$ is constant, $\bar{y} - \mu = \frac{1}{n}\sum_k Z_k(u_k - \mu)$; write $d_k = u_k - \mu$, so $\sum_k d_k = 0$ and $\sum_k d_k^2 = (N-1)S^2$. Then

$$n^2\mathrm{Var}(\bar{y}) = \sum_k d_k^2 \mathrm{Var}Z_k + \sum_{k\neq l} d_k d_l \mathrm{Cov}(Z_k, Z_l) = f(1-f)\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big].$$

Step 4. Because $\sum_{k\neq l} d_k d_l = \big(\sum_k d_k\big)^2 - \sum_k d_k^2 = -\sum_k d_k^2$, the bracket equals $\sum_k d_k^2(1 + \frac{1}{N-1}) = \frac{N}{N-1}(N-1)S^2 = NS^2$. Hence

$$\mathrm{Var}(\bar{y}) = \frac{f(1-f)NS^2}{n^2} = \frac{S^2}{n}(1-f).$$

The same computation with arbitrary inclusion probabilities π_k leads to the Horvitz–Thompson estimator $\sum_k Z_k u_k/\pi_k$ of τ. ■

Exercise 2.2 (Covariance of two sample means)

Each unit carries two values (x_k, y_k), $k = 1, \dots, N$, with population means μ_x, μ_y and

$$S_{xy} = \frac{1}{N-1}\sum_{k=1}^{N}(x_k - \mu_x)(y_k - \mu_y).$$

For an SRS of n pairs without replacement, show that
(a) $\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n} S_{xy}$;
(b) $s_{xy} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$ satisfies $\mathrm{E}(s_{xy}) = S_{xy}$.

Solution. [Supplement] Apply the theorem on the sample mean under SRS to the auxiliary variable $z_k = x_k + y_k$, whose sample mean is $\bar{z} = \bar{x} + \bar{y}$.

(a)

Step 1. Expanding the square, $S_z^2 = \frac{1}{N-1}\sum_k[(x_k - \mu_x) + (y_k - \mu_y)]^2 = S_x^2 + S_y^2 + 2S_{xy}$.

Step 2. By the theorem, $\mathrm{Var}(\bar{z}) = \frac{1-f}{n} S_z^2$, $\mathrm{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$ and $\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2$.

Step 3. Since $\mathrm{Var}(\bar z) = \mathrm{Var}(\bar x) + \mathrm{Var}(\bar y) + 2\mathrm{Cov}(\bar x, \bar y)$,

$$2\mathrm{Cov}(\bar x, \bar y) = \frac{1-f}{n}(S_z^2 - S_x^2 - S_y^2) = \frac{1-f}{n}\,2S_{xy}.$$

(b) The sample analogue of Step 1 is $s_z^2 = s_x^2 + s_y^2 + 2s_{xy}$. Taking expectations and using $\mathrm{E}(s_z^2) = S_z^2$, $\mathrm{E}(s_x^2) = S_x^2$, $\mathrm{E}(s_y^2) = S_y^2$ from the theorem,

$$2\mathrm{E}(s_{xy}) = S_z^2 - S_x^2 - S_y^2 = 2S_{xy}.$$

This polarization argument turns every variance statement about SRS means into a covariance statement; it is the step used for the bias of the ratio estimator in Ch. 5. ■

Exercise 2.3 (Hunters of game birds)

An SRS of $n = 1000$ of the $N = 99{,}000$ licensed hunters of a province is taken, and 430 of them hunt game birds.
(a) Estimate the proportion p of hunters seeking game birds and place a bound on the error of estimation.
(b) How large a sample is needed to estimate p with a bound $B = 0.02$, using the estimate of (a) as prior value?

Solution.

(a) $\hat p = 430/1000 = 0.43$ and, by the proportion theorem,

$$\widehat{\mathrm{Var}}(\hat p) = \frac{\hat p\hat q}{n-1}\cdot\frac{N-n}{N} = \frac{0.43\times 0.57}{999}\cdot\frac{98{,}000}{99{,}000} = 0.000243, \qquad 2\sqrt{0.000243} = 0.031,$$

so p is estimated by 0.43 ± 0.031.

(b) With $D = B^2/4 = 0.0001$ and $pq = 0.43\times 0.57 = 0.2451$,

$$n = \frac{Npq}{(N-1)D + pq} = \frac{99{,}000\times 0.2451}{98{,}999\times 0.0001 + 0.2451} = \frac{24{,}265}{10.145} = 2391.8,$$

so $n = 2392$ hunters are needed. Since N is large, this is close to the infinite-population value $pq/D = 2451$; the fpc saves about 60 interviews. ■